Comparing MLP, LSTM, and Transformer Policies for Robot Imitation Learning on PushT

Project Update | 5 November 2026

UBC EECE 568
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Project update

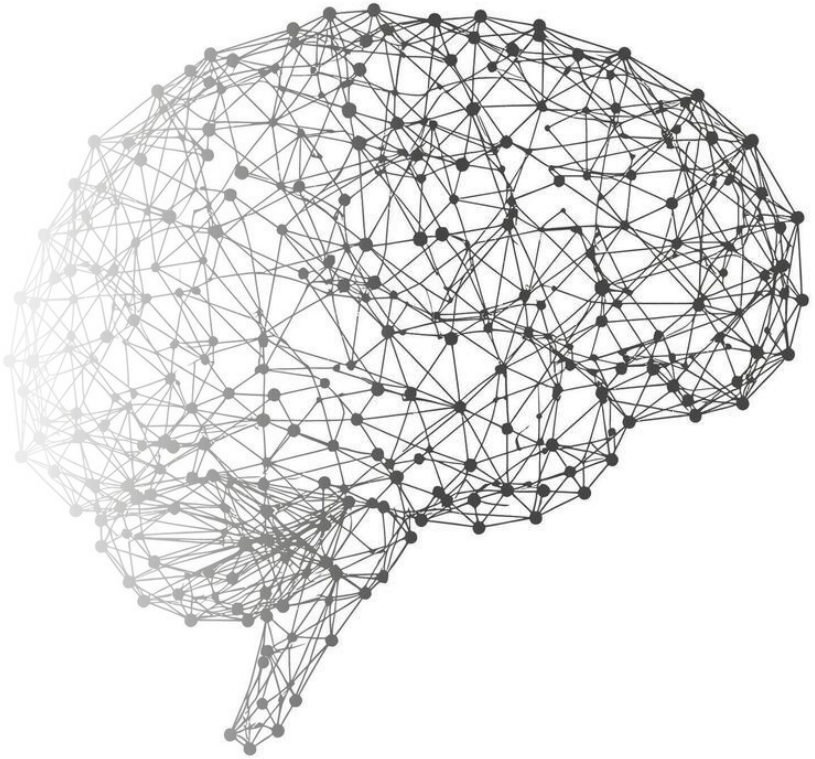
Official keypoint dataset

Episode splits and cross-validation

MLP architecture and training

Saved MLP results and early stopping

Next experiments



Official PushT keypoint dataset

206 episodes • 25,650 frames • 10 Hz

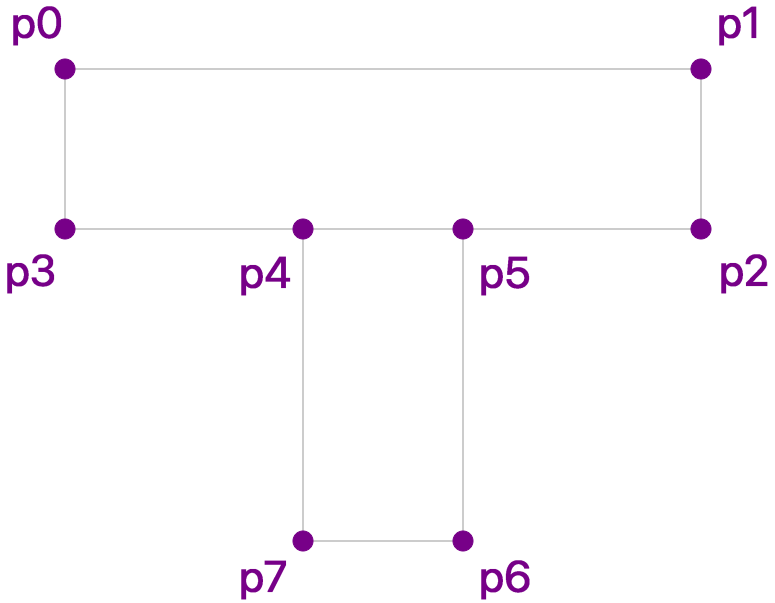


Label: expert pusher target (x, y)

v3 is a storage format; the keypoint variant supplies the extra feature.

Block-state features

Eight 2D landmarks on the rigid T block



Schematic pose; point indices follow gym-pusht.

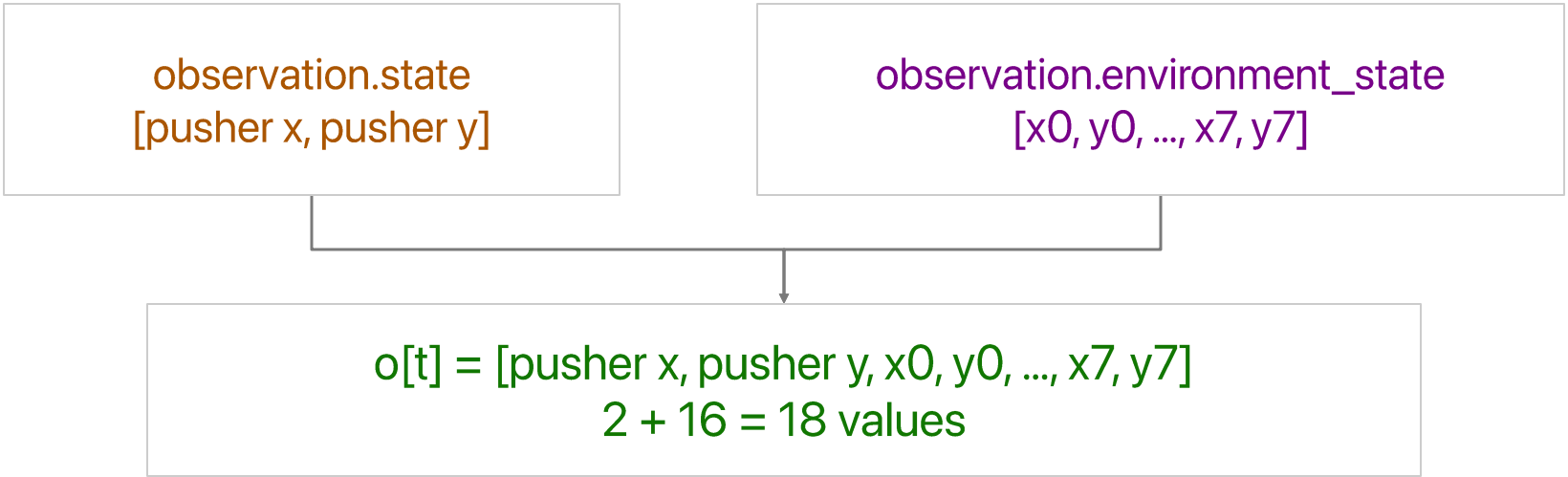
Each landmark contributes an (x, y) pair.

8 points × 2 coordinates
= 16 block-state values

Translation and rotation change these coordinates.

The 18-value observation

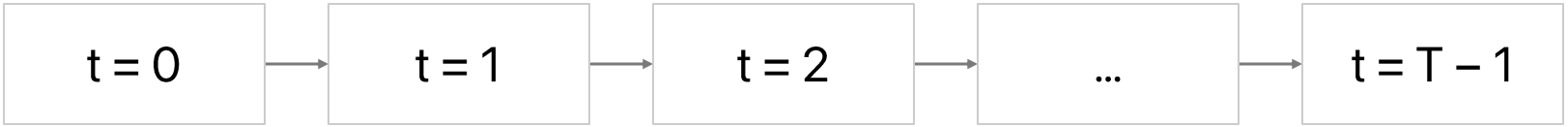
Pusher coordinates followed by block-keypoint coordinates



The block pose can change the needed action at the same pusher position.

Episodes and frames

One episode is one recorded demonstration

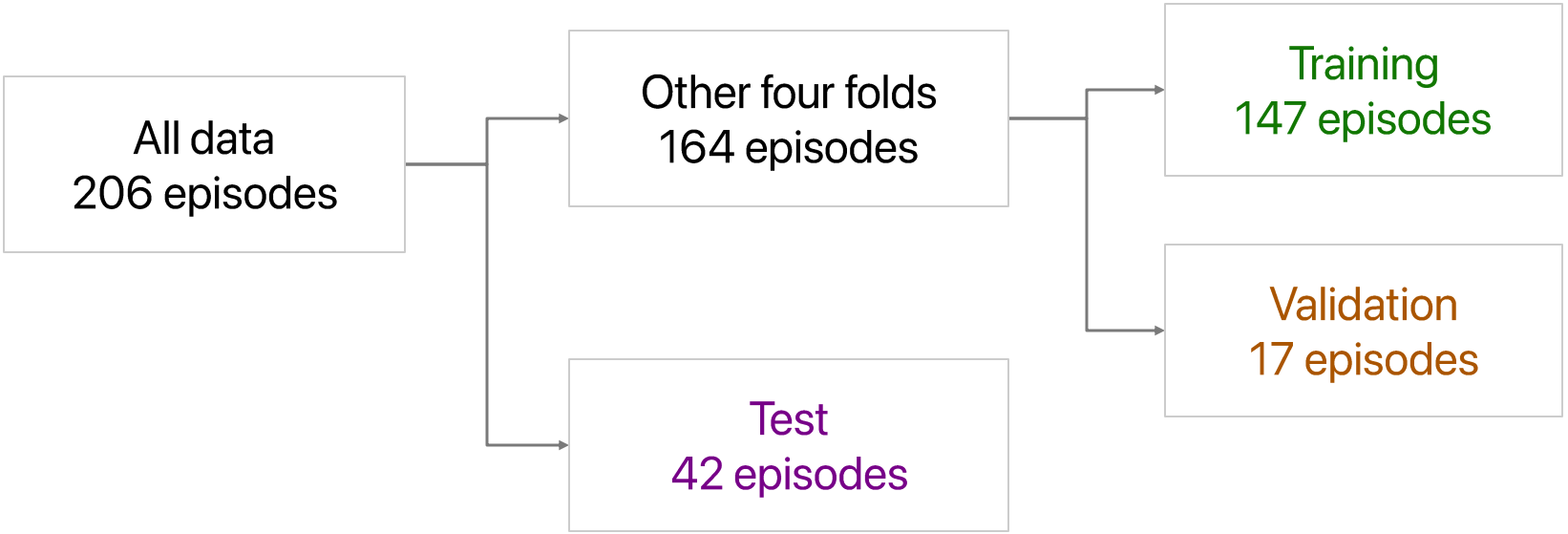


Each frame records:

18D observation
2D expert action
Episode ID, frame ID and timestamp

Episode-based data split

Default experiment: FOLD = 0



18,233 training • 2,147 validation • 5,270 test frames

Five-fold cross-validation

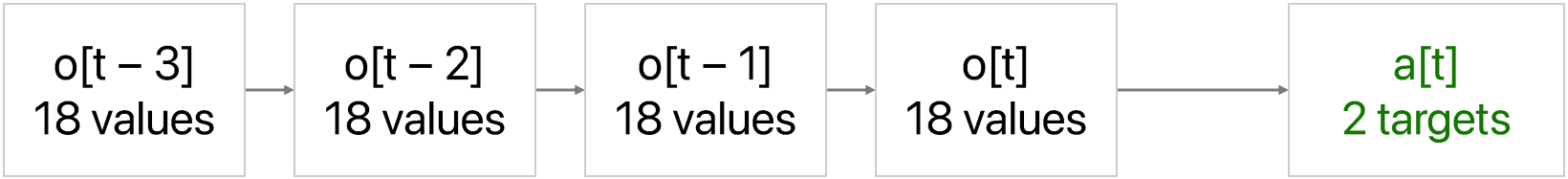
Run	Group A	Group B	Group C	Group D	Group E
FOLD=0	TEST	Pool	Pool	Pool	Pool
FOLD=1	Pool	TEST	Pool	Pool	Pool
FOLD=2	Pool	Pool	TEST	Pool	Pool
FOLD=3	Pool	Pool	Pool	TEST	Pool
FOLD=4	Pool	Pool	Pool	Pool	TEST

Pool = training + validation episodes

Fresh weights and train-only normalization for each run.

Temporal samples and normalization

Default: K = 4 observations → H = 1 action



Normalize inputs and actions with training means and standard deviations.

Repeat the first observation for short histories. Mask unavailable future actions when H = 8.

History and action-chunk settings

K controls context; H controls the output chunk

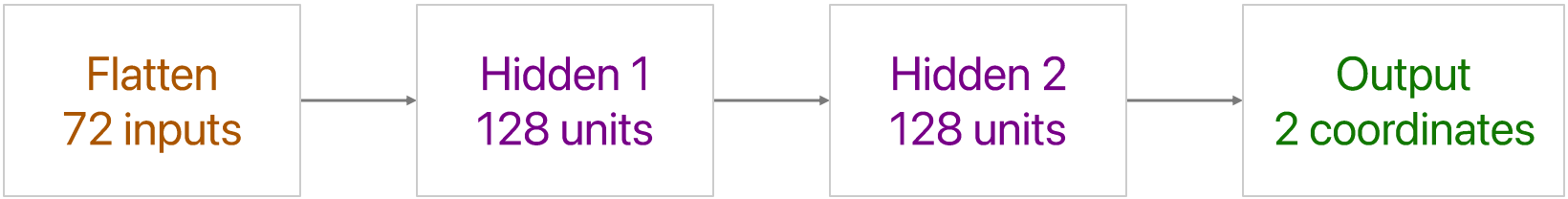
History length K = 4 → o[t−3], o[t−2], o[t−1], o[t]
4 × 18 = 72 input values

Action horizon H = 1 → Predict a[t] = (target x, target y)
H = 8 would predict 16 numbers

Allowed: $K \in \{1, 2, 4, 8\}$; $H \in \{1, 8\}$. Retrain after changing them.

MLP architecture

Default input: 4 × 18 = 72 values



ReLU + dropout 0.1

26,114 trainable parameters • Continuous action regression

Batch size, learning rate and epochs

Three different controls on the training process



An epoch visits every training window once; the order is shuffled.

Batch size affects memory and update noise. Learning rate controls update scale.

Regularization and network capacity

Reduce overfitting without using test data

Dropout = 0.1 → Random activations off in training
All activations available in evaluation

Weight decay = 0.0001 → L2-style penalty in Adam
Discourages large learned weights

Hidden widths 128 / 128 set capacity; they are edited in the Network cell.

Training the MLP

One epoch = one pass through the training windows



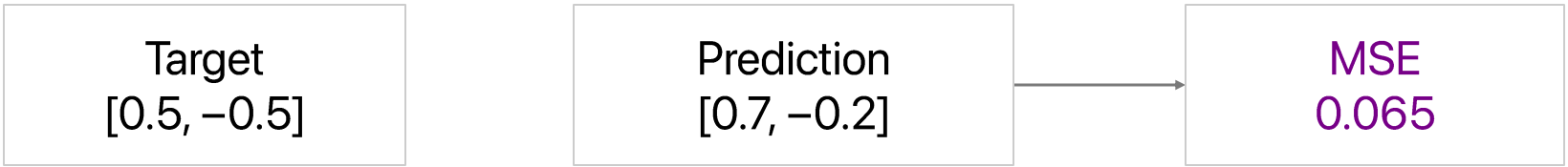
Adam: learning rate 0.001 • batch size 256 • weight decay 0.0001

At most 100 epochs; stop after 3 consecutive epochs without a new best validation MSE.

Masked mean squared error

Continuous x/y targets; square each coordinate error

Loss = total valid squared coordinate error
÷ (2 × number of valid actions)



$(0.2^2 + 0.3^2) \div 2 = 0.065$

Masked loss at an episode boundary

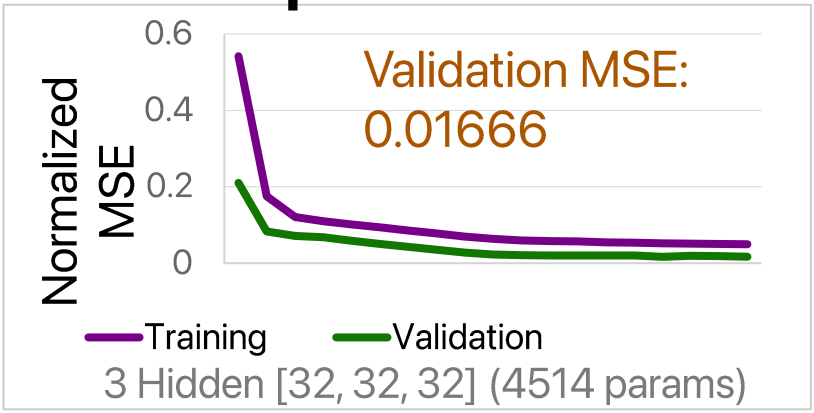
Illustration: H = 8, but only two expert actions remain

a[t]	a[t+1]	a[t+2]	a[t+3]	a[t+4]	a[t+5]	a[t+6]	a[t+7]
valid	valid	padded	padded	padded	padded	padded	padded
1	1	0	0	0	0	0	0

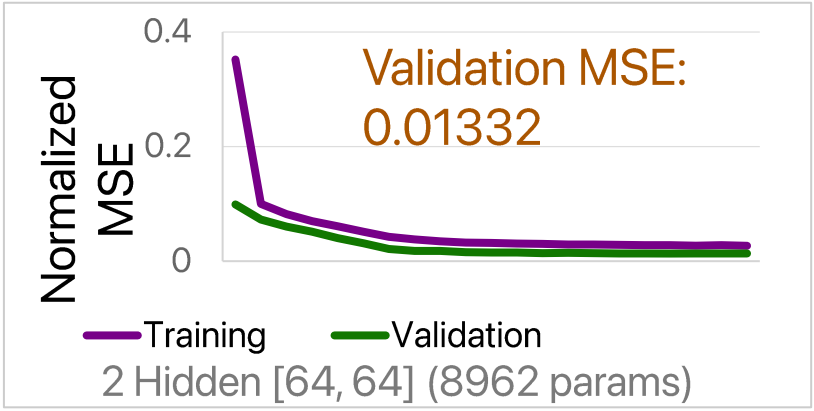
Mask

Only the two valid (x, y) actions contribute: divide by 2 × 2 = 4.

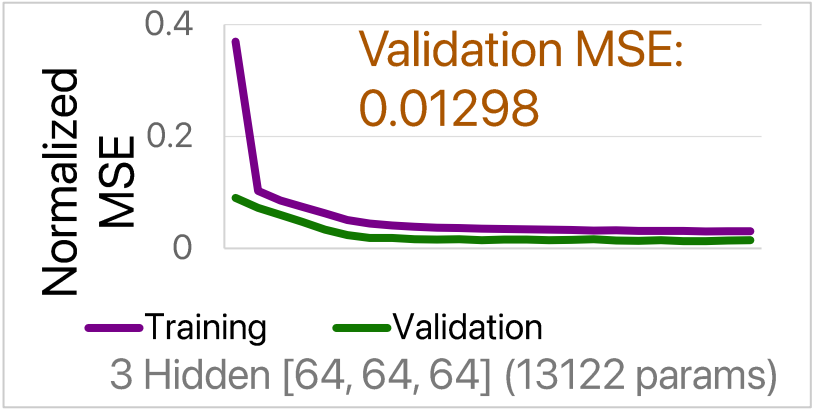
MLP Experimentation



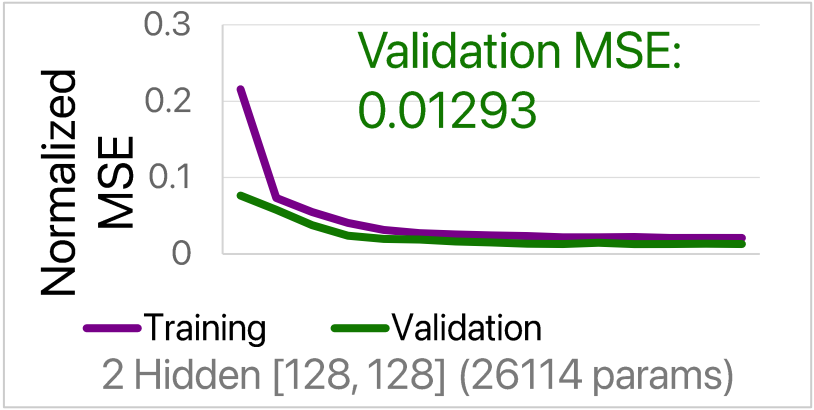
3 Hidden [32, 32, 32] (4514 params)



2 Hidden [64, 64] (8962 params)



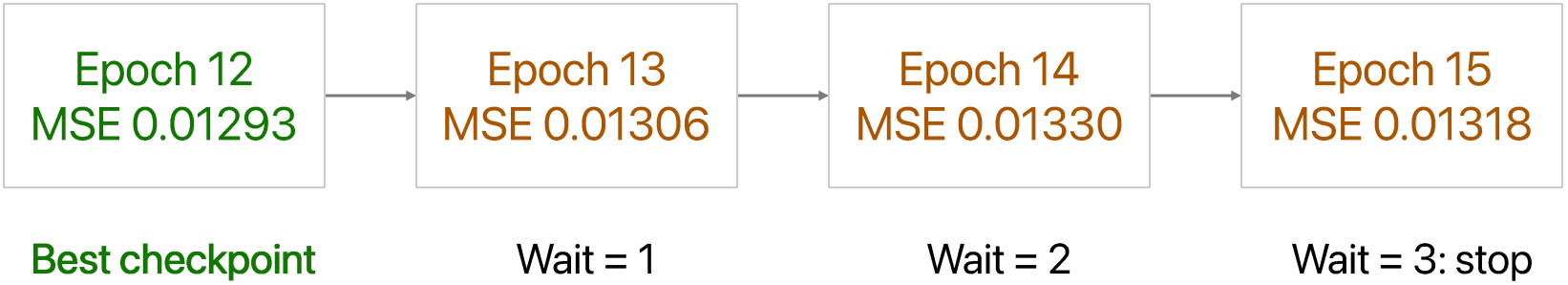
3 Hidden [64, 64, 64] (13122 params)



2 Hidden [128, 128] (26114 params)

Why training stopped at epoch 15

100 was the limit; patience was 3 epochs

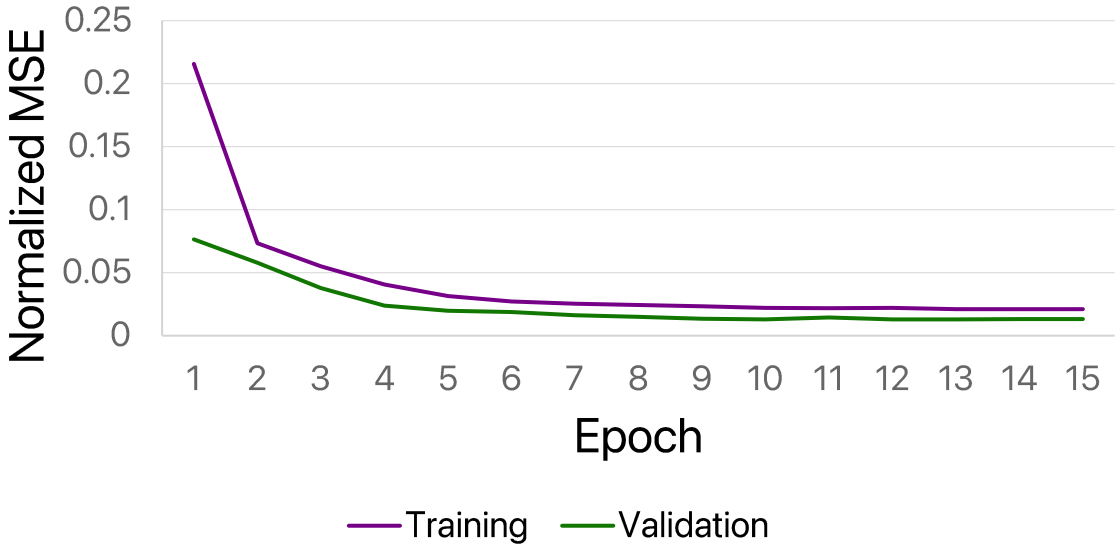


Restore epoch 12 before test evaluation.

Epoch 15 improves on epoch 14, but does not beat the best-ever MSE.

MLP learning curve

Measured run: fold 0 • seed 42 • K = 4 • H = 1



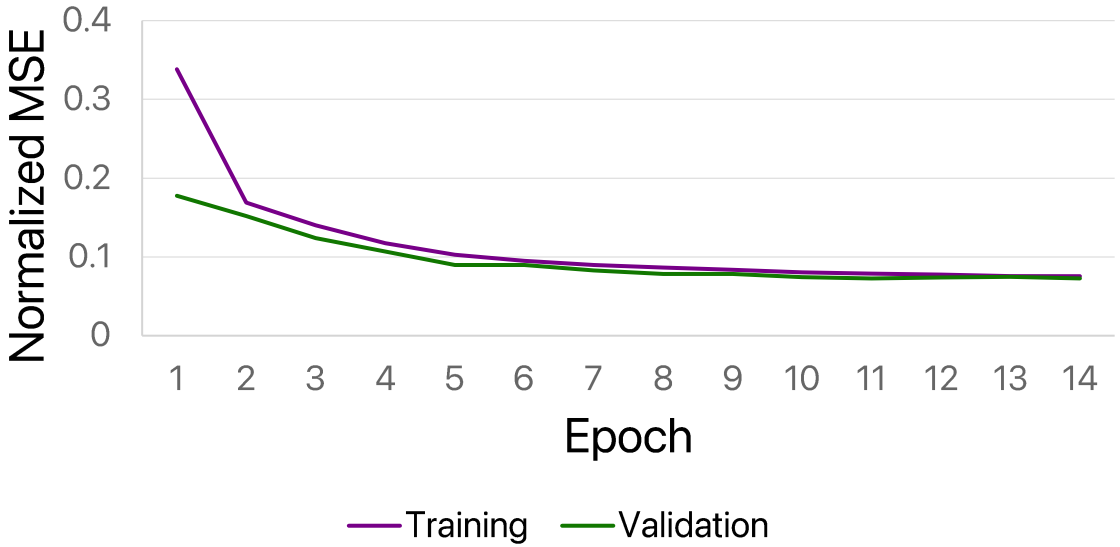
15 epochs run

Best epoch: 12

Validation MSE: 0.01293

MLP learning curve

Measured run: fold 0 • seed 42 • K = 4 • H = 8



14 epochs run

Best epoch: 11

Validation MSE: 0.07266

Held-out offline test

One fold and one training seed; best validation checkpoint

Metric	Saved MLP result
Normalized MSE	0.016212
Raw-coordinate MSE	160.710
Episode-mean raw MSE	158.472

42 unseen episodes • 5,270 test frames

Lower action error is encouraging. Simulator overlap and success have not been tested.

Progress and next experiments

Completed

- Verified 18D keypoint observations and retained episode splits
- Saved the MLP run; early stopping selected epoch 12 ($H = 1$)

Next

- Finish LSTM and then Transformer networks
- Compare histories, action chunks and all three architectures
- Repeat folds/seeds, then test policies in the simulator